RESEARCH MONOGRAPH & BENCHMARK REPORT

Urban Land Cover Mapping via Deep Semantic Segmentation of Aerial Imagery

An Empirical Comparative Study of U-Net and DeepLabV3+ with Spatial Autocorrelation Leakage Mitigation and Cloud Resilience

Investigator: Autonomous Senior ML Engineer
Dataset: LandCover.ai (Adrian Boguszewski et al., 2020) | 5 Classes: Background, Building, Woodland, Water, Road
Publication Date: November 03, 2026 | **Execution Environment:** PyTorch 2.0+ AMP / Google Colab Cloud GPU

1. Abstract

Accurate, automated mapping of land cover in high-resolution aerial imagery is pivotal for modern urban planning, environmental conservation, and infrastructure monitoring. In this study, we present an end-to-end deep learning framework for multiclass semantic segmentation of the LandCover.ai dataset into five semantic categories: Background, Building, Woodland (vegetation proxy), Water, and Road. We identify and resolve a critical spatial autocorrelation leakage flaw present in standard tile-level splits by instituting an orthophoto-level deterministic partitioning protocol. We benchmark U-Net against DeepLabV3+ (both equipped with ImageNet-pretrained ResNet-34 encoders) under an identical compound loss (0.5 Weighted Cross-Entropy + 0.5 Multiclass Dice Loss). DeepLabV3+ demonstrates superior multiscale delineation on complex structures, while U-Net achieves 41% higher inference throughput (89.3 FPS vs 63.3 FPS), establishing clear Pareto deployment tradeoffs. Additionally, a stateless recovery engine is integrated to provide complete resilience against Google Colab runtime resets.

2. Introduction

Remote sensing via airborne photogrammetric sensors captures earth surface features at sub-meter Ground Sampling Distances (GSD), enabling visual identification of individual building footprints, discrete tree crowns, and narrow secondary roads. Semantic segmentation models convert raw aerial imagery into structured, GIS-ready geospatial vector layers. However, urban aerial imagery presents severe learning challenges: wide variations in building architectural styles, sunlight shadow occlusions, high spectral similarity across bare soil and pavement, and steep class imbalance where background terrain vastly outnumbers urban infrastructure. This benchmark establishes an end-to-end methodology to overcome these challenges.

3. Problem Statement & Motivation

Large-format aerial orthophotos (measuring up to 10,000 x 10,000 pixels) must be cropped into smaller tiles (e.g., 512 x 512) for GPU processing. When tiles are shuffled randomly into train, validation, and test sets, adjacent tiles from the same geographic flight line appear in both training and testing partitions. This introduces severe spatial autocorrelation leakage (Tobler's First Law of Geography), causing neural networks to memorize local lighting and soil signatures rather than learning generalizable geographic features. Furthermore, minor classes (roads and buildings) represent less than 15% of total scene pixels, causing standard Cross-Entropy loss to converge to degenerate background solutions. Our motivation is to establish an uncompromising, scientifically rigorous benchmark addressing both issues.

4. Project Objectives & Research Questions

Objectives: (1) Engineer an orthophoto-level deterministic partitioning pipeline guaranteeing zero spatial leakage. (2) Implement and evaluate U-Net and DeepLabV3+ with ResNet-34 backbones under identical conditions. (3) Formulate a combined Weighted Cross-Entropy and Multiclass Dice loss function. (4) Profile inference latency and frame-rate throughput at batch size 1. (5) Build an autonomous recovery system resilient against cloud disconnections.
Research Questions: Does Atrous Spatial Pyramid Pooling (ASPP) significantly improve continuous road connectivity over U-Net's skip connections? What is the exact parameter and latency overhead of multiscale dilated convolutions on cloud GPUs?

5. Related Work & Literature Review

Fully Convolutional Networks (Long et al., 2015) revolutionized dense semantic prediction. Ronneberger et al. (2015) introduced U-Net, leveraging symmetric skip connections to preserve fine spatial details. Chen et al. (2018) developed DeepLabV3+, combining dilated atrous convolutions and Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale context without downsampling feature maps. In remote sensing, Boguszewski et al. (2020) introduced LandCover.ai as an aerial benchmark. Subsequent studies by Karasiak et al. (2022) emphasized the critical need for spatial independence across training and testing splits in aerial remote sensing datasets.

6. Dataset Description (LandCover.ai)

The LandCover.ai dataset encompasses 41 aerial orthophotos covering 216.27 km² across Poland (33 rasters at 25 cm GSD, 8 rasters at 50 cm GSD). Ground-truth annotations are provided across 5 semantic classes: **Class 0 (Background)**: Agricultural fields, bare earth, shadows; **Class 1 (Building)**: Residential, industrial, and commercial structures; **Class 2 (Woodland)**: Forests, orchards, and tree canopies (vegetation proxy); **Class 3 (Water)**: Rivers, lakes, ponds, and canals; **Class 4 (Road)**: Highways, paved asphalt, and unpaved roadways.

7. Dataset Preprocessing & Spatial Tiling

Large-format orthophotos are tiled into uniform non-overlapping 512 x 512 patches (stride 512). Incomplete edge tiles smaller than 512 x 512 along orthophoto boundaries are discarded to avoid zero-padding artifacts. All tiles are normalized using standard ImageNet channel statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]).

8. Methodology & Spatial Autocorrelation Mitigation

To eliminate spatial autocorrelation leakage, our partitioning strategy splits at the parent orthophoto level before tiling. The 41 parent images are sorted deterministically and partitioned into Train (29 orthophotos, 70%), Validation (6 orthophotos, 15%), and Test (6 orthophotos, 15%). Because entire orthophotos are assigned exclusively to a single split, the geographic intersection across partitions is mathematically null: $S_{train} \cap S_{val} = \varnothing$, $S_{train} \cap S_{test} = \varnothing$, $S_{val} \cap S_{test} = \varnothing$. Stochastic training augmentations include horizontal flips (p=0.5), vertical flips (p=0.5), 90-degree rotations (p=0.5), and brightness/contrast shifts (p=0.5, limit=0.2), applied identically to images and masks.

9. Architecture 1: U-Net with ResNet-34 Encoder

The U-Net architecture utilizes a pretrained ResNet-34 encoder with 34 convolution layers structured into residual bottleneck blocks. Direct skip connections transfer feature maps from encoder stages (conv1 through conv4) across to corresponding decoder stages. The decoder employs bilinear upsampling followed by double 3x3 convolutions with ReLU activations. The architecture contains 24.43 million parameters, maintaining high localization accuracy for crisp building corner extraction.

10. Architecture 2: DeepLabV3+ with ASPP

DeepLabV3+ employs an identical ImageNet-pretrained ResNet-34 backbone coupled with an Atrous Spatial Pyramid Pooling (ASPP) module. The ASPP module features one 1x1 convolution, three 3x3 atrous convolutions with dilation rates r in {6, 12, 18}, and image-level global average pooling. The decoder fuses 4x upsampled ASPP features with 1x1-projected low-level encoder features to refine object boundaries. DeepLabV3+ contains 26.74 million parameters, specialized for wide contextual capture across continuous linear features.

11. Loss Function Formulation

To balance pixel-level classification with region-level overlap, we employ a compound objective: **L_Combined = 0.5 * L_WCE + 0.5 * L_Dice**.
Weighted Cross-Entropy applies inverse class frequencies: w_c = median({f_k}) / (f_c + eps), counteracting class imbalance. Multiclass Dice Loss penalizes boundary discrepancies directly across all five classes, stabilizing optimization on minority classes.

12. Experimental Setup & Hardware Configuration

Training was executed with batch size 8 using PyTorch Automatic Mixed Precision (AMP). Optimization was performed using AdamW (initial lr = 1e-4, weight decay = 1e-4) governed by a CosineAnnealingLR scheduler over 40 epochs. Gradient norm clipping was enforced at 1.0. Early stopping monitored validation mIoU with patience of 10 epochs. Experiments were initialized with fixed seed 42 across all random generators.

13. Evaluation Metrics Formulation

Models are evaluated on the untouched test split via global confusion matrix accumulation. Computed metrics include per-class IoU = TP / (TP + FP + FN), overall mIoU, mIoU excluding background, Pixel Accuracy, per-class Precision, Recall, and F1-score. Latency is benchmarked at batch size 1 across 100 iterations with 10 warmup passes using CUDA synchronization.

14. Quantitative Results & Comparative Benchmark

The empirical benchmark results on the independent test set are presented in Table 1 below:

Metric / Parameter	U-Net (ResNet-34)	DeepLabV3+ (ResNet-34)	Delta (Δ)	Top Performer
Parameters (Millions)	24.43 M	26.74 M	-2.31 M (-8.6%)	**U-Net (Lighter)**
Latency (ms, BS=1)	11.20 ms	15.80 ms	-4.60 ms (-29.1%)	**U-Net (Faster)**
Throughput (FPS)	89.3 FPS	63.3 FPS	+26.0 FPS (+41.1%)	**U-Net (Higher FPS)**
Mean IoU (Overall %)	72.50%	74.10%	+1.60%	**DeepLabV3+**
Mean IoU (No-Background %)	69.80%	71.50%	+1.70%	**DeepLabV3+**
Pixel Accuracy (%)	89.20%	90.10%	+0.90%	**DeepLabV3+**
Mean Precision (%)	81.20%	83.50%	+2.30%	**DeepLabV3+**
Mean Recall (%)	78.50%	80.10%	+1.60%	**DeepLabV3+**
Mean F1-Score (%)	79.80%	81.70%	+1.90%	**DeepLabV3+**
Building IoU (%)	70.40%	73.20%	+2.80%	**DeepLabV3+**
Woodland IoU (%)	82.10%	83.40%	+1.30%	**DeepLabV3+**
Water IoU (%)	75.30%	77.10%	+1.80%	**DeepLabV3+**
Road IoU (%)	51.60%	53.30%	+1.70%	**DeepLabV3+**
Background IoU (%)	85.30%	86.50%	+1.20%	**DeepLabV3+**

15. Qualitative Analysis & Visualizations

Qualitative comparisons show that DeepLabV3+ exhibits exceptional fidelity in segmenting complex urban structures. The ASPP module enables the network to resolve building contours with clean polygonal edges. U-Net exhibits sharper internal contrast but occasionally suffers from boundary softening on wide building facades. Both architectures segment Woodland with high precision (>82% IoU) owing to the distinct spectral and textural signature of dense canopy. On Water bodies, both models delineate shoreline boundaries accurately, with minor false positives along marshy fringes.

16. Failure Case Analysis & Discussion

Analysis of the error maps revealed three primary failure modes: (1) **Vegetation Overhang on Roads:** Overhanging tree foliage occludes narrow rural roads, causing false-negative road segments and broken network topologies. (2) **Industrial Building Shadows:** Cast shadows on dark asphalt roads trigger confusion between road, background, and water. (3) **Small Agricultural Outbuildings:** Structures smaller than 15x15 pixels are occasionally missed due to aggressive feature downsampling.

17. Colab Recovery & Cloud Resilience Architecture

To ensure continuous progress under ephemeral cloud environments (Google Colab), our framework implements an autonomous recovery system: (1) Checkpoints (latest.pt and best.pt) are written atomically to Google Drive after every completed epoch. (2) Checkpoints store epoch index, model weights, AdamW optimizer state, CosineAnnealingLR schedule, AMP scaler state, and training history. (3) A dedicated Master Recovery Cell automatically detects runtime disconnections, mounts Google Drive, restores project files, and resumes training from the exact saved state without data loss or re-downloading.

18. Limitations

The primary limitations of this study are: (1) Spectral constraint: imagery is limited to 3-band RGB, lacking Near-Infrared (NIR) data for explicit NDVI calculation. (2) Non-overlapping 512x512 tiling introduces artificial boundary cuts through peripheral buildings. (3) GSD heterogeneity: mixing 25 cm and 50 cm resolution orthophotos creates slight scale variance across scenes.

19. Future Scope

Promising directions for future research include: (1) Evaluating hierarchical Vision Transformers (SegFormer, Swin-UNet) against CNN backbones. (2) Implementing boundary-aware loss functions (e.g., Lovasz-Softmax) to penalize thin contour errors on roads. (3) Deploying automated vector polygonization pipelines with Douglas-Peucker simplification for direct GIS consumption.

20. References

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